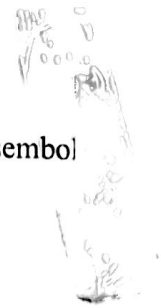
Mazeret Sınavı Cevap Anahtarı

①

ZEMİN A	D_{10}	0,3	C_u	11.67	SP
	D_{30}	0,7	C_c	0.466	
	D_{60}	3,5			
ZEMİN B	D_{10}	0,04	C_u	22,5	SW-SC çift sembol
	D_{30}	0,27	C_c	2,03	
	D_{60}	0,9			
ZEMİN C	D_{10}	0,2	C_u	4,5	SP
	D_{30}	0,45	C_c	1,13	
	D_{60}	0,85			



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② a) $k = \frac{Q.L}{\Delta h.A.t} \rightarrow \frac{15.12}{25.19{,}63.5} = 0.0733\ cm/dk$

$A = \frac{\pi.5^2}{4} = 19.63\ cm^2$

deşarj hızı = $v = k.i \rightarrow 0.0733.\ 25/12 = 0.152\ cm/dk$ ($\hookrightarrow i = \Delta h/L$)

sızıntı hızı = $v_s = v/n \rightarrow \frac{0.152}{0.31} = 0.489\ cm/dk$

$n = e/1+e$

$n = 0.45/1.45 = 0.31$

b) $q = k.\frac{N_f}{N_d}.\Delta H = 0.0733 \cdot 3/6 \cdot 620 = 22.73\ cm^3/dk$

c) $i = \Delta h/L \rightarrow \frac{\Delta H/N_d}{L} = \frac{6.2/6}{1} = 1.033$

$i_c = \gamma'/\gamma_w = \frac{18.5-9.81}{9.81} = 0.885$

$i > i_c \rightarrow$ akıcı kum durumu oluşur.

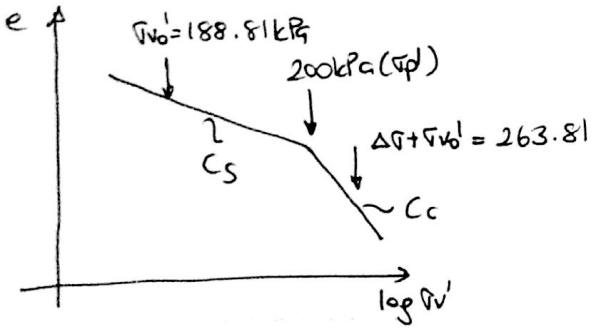
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③ a) $\sigma_{v0}' = 4.17{,}5 + 8.(18-9.81) + 5.5.(19{,}5-9.81) = 188.81\ kPa$

$\Delta\sigma_{v0}' = 75\ kPa$

$\sigma_p' = 200\ kPa$

Aşırı konsolide kil $\rightarrow \sigma_{v0}' + \Delta\sigma_{v0}' = 75 + 188{,}81 = 263.81\ kPa > \sigma_p' = 200\ kPa$



$$\delta = \frac{C_s}{1+e_0}.H.\log\left(\frac{\sigma_p'}{\sigma_{v0}'}\right) + \frac{C_c}{1+e_0}.H.\log\left(\frac{\sigma_{v0}'+\Delta\sigma}{\sigma_p'}\right)$$

$$= \frac{0.06}{1+1.05}.11.\log\left(\frac{200}{188.81}\right) + \frac{0.35}{1+1.05}.11.\log\left(\frac{263.81}{200}\right)$$

$= 0.234\ m \rightarrow \underline{23.4\ cm}$.

b) $U = \%90 \rightarrow T_v = 0.848$

$T_v = \frac{c_v \cdot t}{(H_d)^2} \rightarrow 0.848 = \frac{0.04865 \cdot t}{(1100)^2} \rightarrow t \cong 40\ yıl$

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(4)

a)

1. Deney	2. Deney
$\sigma_3 = 100 kPa$	$\sigma_3 = 200 kPa$
$\sigma_1 - \sigma_3 = 90 kPa$	$\sigma_1 - \sigma_3 = 150 kPa$
$\sigma_1 = 190 kPa$	$\sigma_1 = 350 kPa$

$\sigma_1 = \sigma_3 \tan^2(45+\phi/2) + 2c\tan(45+\phi/2)$

$-/\ 190 = 100 \tan^2(45+\phi'/2) + 2c' \tan(45+\phi'/2)$

$350 = 200 \tan^2(45+\phi'/2) + 2c' \tan(45+\phi'/2)$

$160 = 100 \tan^2(45+\phi'/2)$

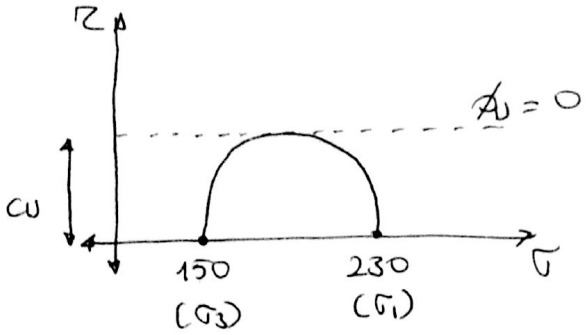
$\boxed{\phi' = 13.34^\circ \quad c' = 11.86 kPa}$

b) $\phi \neq 0$ ve $c \neq 0$ dolayısıyla kil aşırı konsolided

c) $\sigma_3 = 150 kPa$

$\sigma_1 - \sigma_3 = 80 kPa$

$\sigma_1 = 230 kPa$



$\boxed{c_u = 80/2 = 40 kPa \quad \phi_u = 0}$

(5)